

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests all buildings within 1 and 2 km of University of Exeter in Exeter	human, host_agent	Message from human to host_agent: "Show me all buildings within 1 and 2 km of University of Exeter in Exeter"	User initiates the query.
2	Plan the steps to solve the geospatial query	planning_agent	send_message to planning_agent with task_description	planning_agent is chosen to break down the query into actionable steps, as per its system prompt.
3	Planning agent decomposes the query	planning_agent	None (internal reasoning)	Steps are: 1) Find Exeter area, 2) Find University of Exeter in Exeter, 3) Find all buildings in Exeter, 4) Apply condition for buildings within 1 km, 5) Apply condition for buildings within 2 km. This is based on the planning_agent's prompt and the query.
4	Find the area polygon for Exeter	named_area_agent	send_message to named_area with task_description	named_area_agent is chosen because the prompt specifies that named_area contains polygons of areas like cities.
5	Retrieve polygons for Exeter	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	The named_area_agent uses the OS NGD tool to get all named area polygons, as per its API capabilities.
6	Filter to the specific polygon for Exeter	data_analysis_agent	send_message to data_analysis_agent with task_description	Data analysis is needed to filter the correct polygon for Exeter from multiple results, as per the named_area_agent's prompt.
7	Generate metadata for 'exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_area_polygon']	Metadata is generated to understand the columns and structure for accurate filtering.
8	Filter using code executor	coding_agent, code_executor	code_executor with code that filters on 'name1_text', 'description', and 'theme' columns for 'Exeter'	Multiple columns are used for high accuracy, as per the coding agent's guidelines and metadata.
9	Save filtered polygon as 'filtered_exeter_area_polygon'	data_analysis_agent	None (result of code execution)	The filtered result is saved for downstream use.
10	Find the area polygon for University of Exeter in Exeter	named_area_agent	send_message to named_area with task_description	Next step in the plan; named_area_agent is used to find the university polygon.
11	Retrieve polygons for University of Exeter	named_area_agent	call_os_ngd with filters=['University of Exeter'], bbox='filtered_exeter_area_polygon', polygon_or_point=True, filename='university_of_exeter_area_polygon'	Filters and bbox are set to narrow down to the university within Exeter.
12	Filter to the specific polygon for University of Exeter	data_analysis_agent	send_message to data_analysis_agent with task_description	Data analysis is needed to filter the correct university polygon from multiple results.
13	Generate metadata for 'university_of_exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['university_of_exeter_area_polygon']	Metadata is generated to understand the columns and structure for accurate filtering.
14	Filter using code executor	coding_agent, code_executor	code_executor with code that filters on 'name1_text' containing 'Exeter' and 'description' containing 'University'	Multiple columns are used for high accuracy, as per the coding agent's guidelines and metadata.
15	No polygons found for University of Exeter	data_analysis_agent	None (result of code execution)	No matching polygons found; fallback to point location is needed.
16	Find the point location for University of Exeter	address_agent	send_message to address_agent with task_description	Since no polygon found, address_agent is used to find point locations for the university.
17	Retrieve point locations for University of Exeter	address_agent	call_os_ngd with filters=['University'], bbox='filtered_exeter_area_polygon', polygon_or_point=False, filename='university_of_exeter_locations'	Filters and bbox are set to find university addresses within Exeter.
18	Filter to specific point locations for University of Exeter	data_analysis_agent	send_message to data_analysis_agent with task_description	Data analysis is needed to filter the correct university point locations from multiple results.
19	Generate metadata for 'university_of_exeter_locations'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['university_of_exeter_locations']	Metadata is generated to understand the columns and structure for accurate filtering.
20	Filter using code executor	coding_agent, code_executor	code_executor with code that filters on 'organisationname', 'subname', 'name', and 'fulladdress' columns for 'University of Exeter'	Multiple columns are used for high accuracy, as per the coding agent's guidelines and metadata.
21	Save filtered point locations as 'filtered_university_of_exeter_locations'	data_analysis_agent	None (result of code execution)	The filtered result is saved for downstream use.
22	Find all buildings within Exeter	buildings_agent	send_message to buildings_agent with task_description	Next step in the plan; buildings_agent is used to find all buildings within Exeter.
23	Retrieve buildings within Exeter	buildings_agent	call_os_ngd with bbox='filtered_exeter_area_polygon', filename='exeter_buildings'	bbox is set to the Exeter polygon to get all buildings in the area.
24	Save building artifacts	buildings_agent	None (result of tool call)	Artifacts for buildings, building lines, and building parts are generated for further analysis.
25	Plot buildings within 1 km and 2 km of university locations	plotting_agent	send_message to plotting_agent with task_description	plotting_agent is chosen to visualize the results as per the user request and system prompt.
26	Generate metadata for plotting	plotting_agent	generate_metadata_for_artifacts with artifact_names=['buildings_exeter_buildings', 'filtered_university_of_exeter_locations']	Metadata is generated to ensure correct columns and CRS for plotting.
27	Plot using code executor	coding_agent, code_executor	code_executor with code that: buffers university points by 1km and 2km, finds buildings within each buffer, converts CRS, and plots on a folium map	Code uses spatial joins and buffers to find and plot buildings within 1km and 2km, as per plotting agent's guidelines and metadata.
28	Save and return map artifact	plotting_agent	None (result of code execution)	The folium map is saved and returned as the final artifact for user consumption.
29	Present results to user	host_agent	None (summary and map link)	host_agent summarizes the results and offers the interactive map to the user.

Overall Summary: - Databases used: Named area (for Exeter and University of Exeter polygons), Address (for university point locations), Buildings (for all buildings in Exeter). - Filters: Used name and description columns to filter for Exeter and University of Exeter; used bbox to restrict searches to Exeter; used multiple columns for high-accuracy filtering. - Code: Used spatial filtering (buffering, intersection) and CRS transformations to find buildings within 1km and 2km of university locations; plotted results on a folium map. - Final outcome: 15,238 buildings within 1km and 24,693 buildings within 2km of the University of Exeter were identified and visualized on an interactive map. - Overall reasoning: Each agent and tool was chosen based on its system prompt and the requirements of the step (e.g., named_area for polygons, address for points, buildings for building data, plotting_agent for visualization). Filtering and code logic were based on metadata and the need for high-accuracy, multi-column searches. The process followed the planning_agent's decomposition of the query and used spatial analysis best practices for geospatial queries.